1094: METHANOSPHAERULA (PEAT) MEDIUM

Final pH: 5.7
Final volume: 1007 ml

Solution A	943.00	ml
Solution B	13.00	ml
Solution C	20.00	ml
Solution D	1.00	ml
Solution E	10.00	ml
Solution F	10.00	ml
Solution G	10.00	ml

1. Sparge solution A with 80% N_2 and 20% CO_2 gas mixture for 30 - 45 min to make anoxic. Dispense under same gas atmosphere into anoxic Hungate-type tubes or serum vials to 30% of their volume and autoclave. Solutions B, C, D and G are prepared under 100% N_2 gas atmosphere and sterilized by filtration. Solutions E and F are prepared under 100% N_2 gas atmosphere and autoclaved. To complete the medium add appropriate amounts of solutions B to G to the sterile solution A in the sequence indicated and let medium equilibrate overnight. Adjust pH of complete medium to 5.7, if necessary.

2. After inoculation pressurize vials to 1 bar overpressure with sterile 80% H_2 and 20% CO_2 gas mixture.

3. Note: The complete medium is not stable and can be stored only for a short time.

For DSM 25616: Supplement medium with 0.02 g/l yeast extract added to the sterile medium from an anoxic stock solution sterilized by filtration. Adjust pH of complete medium to 6.2. Use 5-10 vol% as inoculum and incubate while shaking at 100 - 200 rpm.

For DSM 25820: Supplement medium with 0.02 g/l yeast extract added to the sterile medium from an anoxic stock solution sterilized by filtration. Use 5-10 vol% as inoculum and incubate while shaking at 100 - 200 rpm.

Solution A

KCl (0.1% w/v)	1.50	ml
KH_2PO_4 (0.1% w/v)	13.60	ml
NH_4Cl (0.1% w/v)	26.80	ml
Trace element solution	1.00	ml
Distilled water	900.00	ml

Solution B

Tris-HCl buffer (1.0 M solution, pH 8.0)	7.20	ml
Nitrilotriacetate (0.5 M, neutralized with 10 N NaOH)	4.80	ml

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$TiCl_3$ (15% w/v solution in HCl; Riedel-de Haen)	0.55	ml

Solution C

2-(N-Morpholino)ethanesulfonic acid (MES)	1.95	g
NaOH	0.20	g
Distilled water	20.00	ml

Adjust to pH 7.5 with NaOH.

Solution D

Wolin's vitamin solution (10x)	1.00	ml

Solution E

2-Mercaptoethanesulfonate (Coenzyme M)	0.08	g
Distilled water	10.00	ml

Solution F

Na-acetate	0.41	g
Distilled water	10.00	ml

Solution G

Na_2S x 9 H_2O	9.60	mg
Distilled water	10.00	ml

Trace element solution

Na_2-EDTA	37.23	g
$FeCl_2$ x 4 H_2O	1.34	g
$AlK(SO_4)_2$ x 12 H_2O	3.45	g
$CaCl_2$ x 2 H_2O	2.34	g
$MgSO_4$ x 7 H_2O	1.56	g
$CoCl_2$ x 6 H_2O	24.00	mg
$ZnCl_2$	75.00	mg
H_3BO_3	19.00	mg
$NiCl_2$ x 6 H_2O	24.00	mg
Na_2Mo_4 x 2 H_2O	24.00	mg
$MnSO_4$ x 4 H_2O	26.00	mg
$CuSO_4$ x 5 H_2O	9.00	mg
Distilled water	1000.00	ml

Dissolve EDTA in distilled water, adjust pH to 7 using 2 N NaOH and add ferrous chloride.

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After ferrous chloride has dissolved add other compounds.

Wolin's vitamin solution (10x) (from medium 120)

Biotin	20.00	mg
Folic acid	20.00	mg
Pyridoxine hydrochloride	100.00	mg
Thiamine HCl	50.00	mg
Riboflavin	50.00	mg
Nicotinic acid	50.00	mg
Calcium D-(+)-pantothenate	50.00	mg
Vitamin B_{12}	1.00	mg
p-Aminobenzoic acid	50.00	mg
(DL)-alpha-Lipoic acid	50.00	mg
Distilled water	1000.00	ml

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